Quantitative Comparisons

ANSWER KEY

Numeric comparisons

1 Column A: 7×8. Column B: $6 \times 9 + 2$. Compare the two columns: which is greater, are they equal, or is it impossible to determine?

$A = 56$. $B = 54 + 2 = 56$. $A = B$.

2 Column A: $123 + 456$. Column B: $700 - 120$. Compare the two columns.

$A = 579$. $B = 580$. $B > A$.

3 Column A: 9^2. Column B: $8^2 + 15$. Compare the two columns.

$A = 81$. $B = 64 + 15 = 79$. $A > B$.

4 Column A: $(-3)^2$. Column B: -3^2. Compare the two columns.

$A = (-3) \times (-3) = 9$. $B = -(3^2) = -9$ (no parentheses around the base). $A > B$.

Fraction and decimal comparisons

5 Column A: $\frac{3}{7}$. Column B: 0.43. Compare the two columns.

$\frac{3}{7} \approx 0.4286$, which is less than 0.43. $B > A$.

6 Column A: $\frac{7}{12}$. Column B: 0.58. Compare the two columns.

$\frac{7}{12} \approx 0.5833$, which is greater than 0.58. $A > B$.

7 Column A: $2\frac{3}{8}$. Column B: $\frac{19}{8}$. Compare the two columns.

$2\frac{3}{8} = \frac{19}{8}$. $A = B$.

8 Column A: 0.6. Column B: $\frac{5}{9}$. Compare the two columns.

$\frac{5}{9} \approx 0.5556$, which is less than 0.6. $A > B$.

Percent comparisons

9 Column A: 25% of 160. Column B: 20% of 200. Compare the two columns.

$A = 0.25 \times 160 = 40$. $B = 0.20 \times 200 = 40$. $A = B$.

10 Column A: 30% of 90. Column B: 40% of 60. Compare the two columns.

$A = 0.3 \times 90 = 27$. $B = 0.4 \times 60 = 24$. $A > B$.

11 Column A: the result of increasing 100 by 10% twice in a row. Column B: the result of increasing 100 by 20% once. Compare the two columns.

$A = 100 \times 1.1 \times 1.1 = 121$. $B = 100 \times 1.2 = 120$. $A > B$.

12 Column A: 15% of 80. Column B: 12% of 100. Compare the two columns.
$A = 0.15 \times 80 = 12$. $B = 0.12 \times 100 = 12$. $A = B$.

Exponent and root comparisons

13 Column A: $\sqrt{50}$. Column B: 7.1. Compare the two columns.
$\sqrt{50} \approx 7.07$, which is less than 7.1. $B > A$.

14 Column A: 2^{10}. Column B: 10^3. Compare the two columns.
$A = 1{,}024$. $B = 1{,}000$. $A > B$.

15 Column A: $\sqrt{0.09}$. Column B: 0.3. Compare the two columns.
$\sqrt{0.09} = 0.3$. $A = B$.

16 Column A: 3^4. Column B: 4^3. Compare the two columns.
$A = 81$. $B = 64$. $A > B$.

Algebraic comparisons at a fixed value

17 Given $x = 5$. Column A: $2x + 3$. Column B: $x + 8$. Compare the two columns.
$A = 2(5) + 3 = 13$. $B = 5 + 8 = 13$. $A = B$.

18 Given $x = 4$. Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.
$(x + 2)^2$ expands to $x^2 + 4x + 4$ for every value of x, so the columns are always equal. $A = B$.

19 Given $n = -2$. Column A: n^2. Column B: n^3. Compare the two columns.
$A = (-2)^2 = 4$. $B = (-2)^3 = -8$. $A > B$.

20 Given $x = 3$. Column A: $5x - 2$. Column B: $2(x + 5)$. Compare the two columns.
$A = 5(3) - 2 = 13$. $B = 2(3 + 5) = 16$. $B > A$.

Comparisons requiring testing values

21 For any real number x: Column A: x. Column B: x^2. Compare the two columns.
If $x = 2$: $B > A$. If $x = \frac{1}{2}$: $A > B$. If $x = 1$: $A = B$. The relationship changes, so it cannot be determined.

22 For any real number x: Column A: $x + 5$. Column B: $2x$. Compare the two columns.
If $x = 5$: $A = B$. If $x = 0$: $A > B$. If $x = 10$: $B > A$. The relationship changes, so it cannot be determined.

23 For any real number x: Column A: $|x|$. Column B: x. Compare the two columns.

If $x \geq 0$: $A = B$. If $x < 0$ (e.g. $x = -3$): $A = 3$, $B = -3$, so $A > B$. Since the relationship is not the same in every case, it cannot be determined.

24 Given only that n is an integer: Column A: n^2. Column B: 4. Compare the two columns.

If $n = 1$: $A = 1 < 4$. If $n = 3$: $A = 9 > 4$. The relationship changes, so it cannot be determined.

25 For any real number x: Column A: $(x-3)^2$. Column B: $x^2 - 6x + 9$. Compare the two columns.

Expanding: $(x-3)^2 = x^2 - 6x + 9$ for every value of x, so the columns are always equal. $A = B$.

Geometry comparisons

26 Column A: the perimeter of a square with side 6 cm. Column B: the perimeter of a rectangle with length 8 cm and width 3 cm. Compare the two columns.

$A = 4 \times 6 = 24$ cm. $B = 2(8+3) = 22$ cm. $A > B$.

27 Column A: the area of a circle with radius 4 cm (use $\pi \approx 3.14$). Column B: the area of a square with side 7 cm. Compare the two columns.

$A = 3.14 \times 4^2 = 3.14 \times 16 = 50.24$ cm^2. $B = 7^2 = 49$ cm^2. $A > B$.

28 Column A: the sum of the interior angles of a pentagon. Column B: the sum of the interior angles of two triangles. Compare the two columns.

$A = (5-2) \times 180^\circ = 540^\circ$. $B = 2 \times 180^\circ = 360^\circ$. $A > B$.

29 Column A: the hypotenuse of a right triangle with legs 5 and 12. Column B: 14. Compare the two columns.

$A = \sqrt{5^2 + 12^2} = \sqrt{169} = 13$. $B = 14$. $B > A$.

Ratio and proportion comparisons

30 Column A: the value of x, if $x : 12 = 3 : 4$. Column B: 10. Compare the two columns.

$x = \dfrac{3 \times 12}{4} = 9$. $B = 10$. $B > A$.

31 A recipe uses flour and sugar in the ratio $3 : 1$. Column A: the sugar needed for 240 g of flour. Column B: 90 g. Compare the two columns.

Sugar $= 240 \div 3 = 80$ g. $B = 90$ g. $B > A$.

32 Column A: the sum of the parts in the simplified form of the ratio $18 : 24$. Column B: 8. Compare the two columns.

$18 : 24$ simplifies to $3 : 4$, so the sum of parts is $3 + 4 = 7$. $B = 8$. $B > A$.

33 Two numbers are in the ratio $5:3$, and their sum is 96. Column A: the larger number. Column B: 61. Compare the two columns.

Total parts $= 5 + 3 = 8$, each part $= 12$. Larger number $= 5 \times 12 = 60$. $B = 61$. $B > A$.

Data and probability comparisons

34 A data set is $4, 7, 7, 9, 13$. Column A: the mean. Column B: the median. Compare the two columns.

Mean $= \frac{4 + 7 + 7 + 9 + 13}{5} = \frac{40}{5} = 8$. Median (middle value) $= 7$. $A > B$.

35 Column A: the range of the data set $3, 8, 15, 22$. Column B: 20. Compare the two columns.

Range $= 22 - 3 = 19$. $B = 20$. $B > A$.

36 A bag contains 3 red balls and 5 blue balls. Column A: the probability of drawing a red ball. Column B: the probability of drawing a blue ball. Compare the two columns.

$P(\text{red}) = \frac{3}{8}$. $P(\text{blue}) = \frac{5}{8}$. $B > A$.

Word-problem comparisons

37 Column A: the total cost of 4 items at 15 riyals each, after a 10% discount on the total. Column B: 55 riyals. Compare the two columns.

Total before discount $= 4 \times 15 = 60$ riyals. After discount: $60 \times 0.9 = 54$ riyals. $B = 55$. $B > A$.

38 Ali's age is 3 years more than twice Sara's age. Sara is 8. Column A: Ali's age. Column B: 20. Compare the two columns.

Ali $= 2(8) + 3 = 19$. $B = 20$. $B > A$.

39 Column A: the simple interest earned on 2,000 riyals at 5% per year for 3 years. Column B: 350 riyals. Compare the two columns.

$I = 2{,}000 \times 0.05 \times 3 = 300$ riyals. $B = 350$. $B > A$.

40 A car travels 240 km using 20 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km per liter. Compare the two columns.

Distance per liter $= 240 \div 20 = 12$ km/liter. $B = 12$. $A = B$.

Comparisons with inequalities and ranges

41 Given that $5 < x < 9$. Column A: x. Column B: 7. Compare the two columns.

If $x = 6$: $A < B$. If $x = 8$: $A > B$. The relationship changes, so it cannot be determined.

42 Given that $x > 10$. Column A: x. Column B: 10. Compare the two columns.

Since x is always greater than 10, $A > B$ in every case.

43 Given that $0 < x < 1$. Column A: x. Column B: x^2. Compare the two columns.

For any value strictly between 0 and 1 (e.g. $x = 0.5$: $x^2 = 0.25 < 0.5$), squaring makes the value smaller. $A > B$ in every case.

44 Given that $-5 \leq x \leq -1$. Column A: x^2. Column B: 10. Compare the two columns.

If $x = -1$: $A = 1 < 10$. If $x = -5$: $A = 25 > 10$. The relationship changes, so it cannot be determined.

More geometry comparisons

45 Column A: the volume of a cube with edge 4 cm. Column B: the volume of a rectangular box $3 \text{ cm} \times 4 \text{ cm} \times 6 \text{ cm}$. Compare the two columns.

$A = 4^3 = 64 \text{ cm}^3$. $B = 3 \times 4 \times 6 = 72 \text{ cm}^3$. $B > A$.

46 Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the perimeter of a square with side 11 cm. Compare the two columns.

$A = 2 \times \frac{22}{7} \times 7 = 44$ cm. $B = 4 \times 11 = 44$ cm. $A = B$.

47 Column A: the area of a triangle with base 10 and height 8. Column B: the area of a rectangle with length 9 and width 5. Compare the two columns.

$A = \frac{1}{2} \times 10 \times 8 = 40$. $B = 9 \times 5 = 45$. $B > A$.

48 Column A: the sum of the interior angles of a quadrilateral. Column B: three times the sum of the interior angles of a triangle. Compare the two columns.

$A = 360^\circ$. $B = 3 \times 180^\circ = 540^\circ$. $B > A$.

MCQ — Numeric comparisons

49 Column A: 6×7. Column B: $5 \times 9 - 3$. Compare the two columns.

a. The two columns are equal ✓

b. Column B is greater

c. The relationship cannot be determined from the given information

d. Column A is greater

42 vs $45 - 3 = 42$: equal.

50 Column A: $345 + 678$. Column B: $900 + 100$. Compare the two columns.

a. **Column A is greater** ✓

b. The relationship cannot be determined from the given information

c. Column B is greater

d. The two columns are equal

$1{,}023$ vs $1{,}000$: Column A is greater.

51 Column A: 11^2. Column B: $10^2 + 23$. Compare the two columns.

a. **Column B is greater** ✓

b. Column A is greater

c. The relationship cannot be determined from the given information

d. The two columns are equal

121 vs 123: Column B is greater.

52 Column A: $(-4)^2$. Column B: -4^2. Compare the two columns.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column B is greater

d. **Column A is greater** ✓

16 vs -16: Column A is greater.

MCQ — Fraction and decimal comparisons

53 Column A: $\frac{5}{8}$. Column B: 0.62. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. The two columns are equal

d. **Column A is greater ✓**

$\frac{5}{8} = 0.625$, which is greater than 0.62.

54 Column A: $\frac{9}{16}$. Column B: 0.5625. Compare the two columns.

a. Column A is greater

b. Column B is greater

c. **The two columns are equal ✓**

d. The relationship cannot be determined from the given information

$\frac{9}{16} = 0.5625$ exactly: equal.

55 Column A: $3\frac{1}{6}$. Column B: $\frac{19}{6}$. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. Column A is greater

d. **The two columns are equal ✓**

$3\frac{1}{6} = \frac{19}{6}$: equal.

56 Column A: 0.7. Column B: $\frac{5}{7}$. Compare the two columns.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. **Column B is greater ✓**

d. Column A is greater

$\frac{5}{7} \approx 0.714$, which is greater than 0.7.

MCQ — Percent comparisons

57 Column A: 20% of 150. Column B: 25% of 120. Compare the two columns.

- **a.** **The two columns are equal ✓**
- **b.** Column A is greater
- **c.** Column B is greater
- **d.** The relationship cannot be determined from the given information

30 vs 30: equal.

58 Column A: 45% of 80. Column B: 40% of 100. Compare the two columns.

- **a.** The relationship cannot be determined from the given information
- **b.** The two columns are equal
- **c.** Column A is greater
- **d.** **Column B is greater ✓**

36 vs 40: Column B is greater.

59 Column A: the result of decreasing 100 by 20%, then increasing that result by 20%. Column B: 96.

- **a.** The relationship cannot be determined from the given information
- **b.** **The two columns are equal ✓**
- **c.** Column A is greater
- **d.** Column B is greater

$100 \times 0.8 \times 1.2 = 96$: equal.

60 Column A: 18% of 50. Column B: 9% of 90. Compare the two columns.

a. Column B is greater

b. The relationship cannot be determined from the given information

c. **Column A is greater ✓**

d. The two columns are equal

9 vs 8.1: Column A is greater.

MCQ — Exponent and root comparisons

61 Column A: $\sqrt{72}$. Column B: 8.5. Compare the two columns.

a. **Column B is greater ✓**

b. Column A is greater

c. The two columns are equal

d. The relationship cannot be determined from the given information

$\sqrt{72} \approx 8.49$, which is less than 8.5.

62 Column A: 3^5. Column B: 5^3. Compare the two columns.

a. **Column A is greater ✓**

b. The two columns are equal

c. Column B is greater

d. The relationship cannot be determined from the given information

243 vs 125: Column A is greater.

63 Column A: $\sqrt{0.16}$. Column B: 0.4. Compare the two columns.

a. Column A is greater

b. Column B is greater

c. The relationship cannot be determined from the given information

d. **The two columns are equal ✓**

$\sqrt{0.16} = 0.4$ exactly: equal.

64 Column A: 2^6. Column B: 6^2. Compare the two columns.

a. The relationship cannot be determined from the given information

b. **Column A is greater ✓**

c. Column B is greater

d. The two columns are equal

64 vs 36: Column A is greater.

MCQ — Algebraic comparisons at a fixed value

65 Given $x = 6$. Column A: $3x - 4$. Column B: $x + 10$. Compare the two columns.

a. **Column B is greater ✓**

b. Column A is greater

c. The relationship cannot be determined from the given information

d. The two columns are equal

14 vs 16: Column B is greater.

66 Given $x = 3$. Column A: $(x + 1)^2$. Column B: $x^2 + 2x + 1$. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. Column A is greater

d. **The two columns are equal** ✓

Both equal 16 (this is an identity, true for every x): equal.

67 Given $n = -3$. Column A: n^2. Column B: n^3. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. The two columns are equal

d. **Column A is greater** ✓

9 vs -27: Column A is greater.

68 Given $x = 2$. Column A: $4x + 1$. Column B: $3(x + 2)$. Compare the two columns.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. **Column B is greater** ✓

9 vs 12: Column B is greater.

MCQ — Comparisons requiring testing values

69 For any real number x: Column A: x^2. Column B: 0. Compare the two columns.

a. The relationship cannot be determined from the given information ✓

b. The two columns are equal

c. Column A is greater

d. Column B is greater

x^2 is always ≥ 0, but could equal 0 (at $x = 0$) or exceed it — not always the same relationship.

70 For any integer n: Column A: n. Column B: $-n$. Compare the two columns.

a. The two columns are equal

b. The relationship cannot be determined from the given information ✓

c. Column B is greater

d. Column A is greater

If $n > 0$, A is greater; if $n < 0$, B is greater; if $n = 0$, they are equal — it varies.

71 For any real number x with $x \neq 0$: Column A: x^2. Column B: 0. Compare the two columns.

a. Column B is greater

b. Column A is greater ✓

c. The two columns are equal

d. The relationship cannot be determined from the given information

Since $x \neq 0$, x^2 is always strictly positive, so Column A is always greater.

72 For any real number x: Column A: $(x + 2)^2$. Column B: $x^2 + 4x + 4$. Compare the two columns.

a. The two columns are equal ✓

b. Column A is greater

c. The relationship cannot be determined from the given information

d. Column B is greater

$(x + 2)^2 = x^2 + 4x + 4$ for every x: equal.

73 Given only that m is a positive integer: Column A: m. Column B: m^2. Compare the two columns.

a. Column B is greater

b. Column A is greater

c. The two columns are equal

d. The relationship cannot be determined from the given information ✓

At $m = 1$ they are equal; for $m > 1$, Column B is greater — it varies.

74 For any real number x: Column A: $-x$. Column B: $x - 1$. Compare the two columns.

a. The two columns are equal

b. Column A is greater

c. The relationship cannot be determined from the given information ✓

d. Column B is greater

$-x - (x - 1) = 1 - 2x$, whose sign depends on x — it varies.

MCQ — Geometry comparisons

75 Column A: the circumference of the circle shown (use $\pi \approx 3.14$). Column B: 32.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. Column B is greater ✓

$C = 2(3.14)(5) = 31.4$, which is less than 32.

76 Column A: the area of the rectangle shown. Column B: the perimeter of a square with side 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column B is greater

d. Column A is greater ✓

Rectangle area $= 9 \times 4 = 36$; square perimeter $= 4 \times 7 = 28$. Column A is greater.

77 Column A: the sum of the interior angles of a hexagon. Column B: the sum of the interior angles of two triangles.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater ✓

d. Column B is greater

Hexagon $= 720^\circ$; two triangles $= 180 + 180 = 360^\circ$. Column A is greater.

78 Column A: the hypotenuse of the right triangle shown. Column B: 16.

a. The relationship cannot be determined from the given information

b. Column B is greater ✓

c. The two columns are equal

d. Column A is greater

$\sqrt{9^2 + 12^2} = \sqrt{225} = 15$, which is less than 16.

79 Column A: the volume of a cylinder with radius 3 and height 7 (use $\pi \approx 3.14$). Column B: 200.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. The two columns are equal

d. Column B is greater ✓

$V = 3.14 \times 9 \times 7 = 197.82$, which is less than 200.

MCQ — Ratio and proportion comparisons

80 Column A: the value of x, if $x : 15 = 2 : 5$. Column B: 7.

a. The relationship cannot be determined from the given information

b. The two columns are equal

c. Column A is greater

d. **Column B is greater ✓**

$x = 15 \times \frac{2}{5} = 6$, which is less than 7.

81 Two numbers are in the ratio $4 : 7$, and their sum is 99. Column A: the smaller number. Column B: 40.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **Column B is greater ✓**

Each part $= 99 \div 11 = 9$; smaller $= 4 \times 9 = 36$, which is less than 40.

82 Column A: the sum of the parts in the simplified form of the ratio $24 : 36$. Column B: 5.

a. Column B is greater

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **The two columns are equal ✓**

$24 : 36$ simplifies to $2 : 3$; $2 + 3 = 5$: equal.

83 A recipe uses water and rice in the ratio $2:1$. Column A: the water needed for 300 g of rice. Column B: 550 g.

a. Column B is greater

b. **Column A is greater ✓**

c. The two columns are equal

d. The relationship cannot be determined from the given information

Water $= 2 \times 300 = 600$ g, which is greater than 550 g.

MCQ — Data and probability comparisons

84 A data set is $5, 8, 8, 10, 14$. Column A: the mean. Column B: the median.

a. The two columns are equal

b. Column B is greater

c. **Column A is greater ✓**

d. The relationship cannot be determined from the given information

Mean $= \frac{45}{5} = 9$; median $= 8$. Column A is greater.

85 Column A: the range of the data set $6, 11, 19, 25$. Column B: 19.

a. The relationship cannot be determined from the given information

b. Column B is greater

c. **The two columns are equal ✓**

d. Column A is greater

Range $= 25 - 6 = 19$: equal.

86 A bag contains 4 green balls and 9 yellow balls. Column A: the probability of drawing a green ball. Column B: 0.4.

a. **Column B is greater ✓**

b. The relationship cannot be determined from the given information

c. The two columns are equal

d. Column A is greater

$P = \frac{4}{13} \approx 0.31$, which is less than 0.4: Column B is greater.

87 A data set is $2, 4, 4, 4, 6$. Column A: the mode. Column B: the mean.

a. Column A is greater

b. The relationship cannot be determined from the given information

c. **The two columns are equal ✓**

d. Column B is greater

Mode $= 4$; mean $= \frac{20}{5} = 4$: equal.

MCQ — Word-problem comparisons

88 Column A: the total cost of 5 items at 20 riyals each, after a 15% discount on the total. Column B: 90.

a. The two columns are equal

b. Column A is greater

c. The relationship cannot be determined from the given information

d. **Column B is greater ✓**

$5 \times 20 = 100$; after discount $= 100 \times 0.85 = 85$, which is less than 90.

89 Layla's age is 4 years less than three times Huda's age. Huda is 9. Column A: Layla's age. Column B: 25.

a. Column A is greater

b. The two columns are equal

c. **Column B is greater ✓**

d. The relationship cannot be determined from the given information

Layla $= 3(9) - 4 = 23$, which is less than 25.

90 Column A: the simple interest on 3,000 riyals at 4% per year for 2 years. Column B: 250.

a. The two columns are equal

b. The relationship cannot be determined from the given information

c. Column A is greater

d. **Column B is greater ✓**

$I = 3{,}000 \times 0.04 \times 2 = 240$, which is less than 250.

91 A car travels 300 km using 25 liters of fuel. Column A: the distance travelled per liter. Column B: 12 km.

a. **The two columns are equal ✓**

b. The relationship cannot be determined from the given information

c. Column B is greater

d. Column A is greater

$300 \div 25 = 12$ km per liter: equal.

MCQ — Comparisons with inequalities and ranges

92 Given that $3 < x < 8$. Column A: x. Column B: 5.

a. The relationship cannot be determined from the given information ✓

b. The two columns are equal

c. Column A is greater

d. Column B is greater

x could be less than, equal to, or greater than 5 within this range — it varies.

93 Given that $x < -4$. Column A: x. Column B: -4.

a. Column A is greater

b. Column B is greater ✓

c. The two columns are equal

d. The relationship cannot be determined from the given information

Every value less than -4 is smaller than -4, so Column B is always greater.

94 Given that $-3 \leq x \leq 2$. Column A: x^2. Column B: 9.

a. Column B is greater

b. Column A is greater

c. The two columns are equal

d. The relationship cannot be determined from the given information ✓

At $x = -3$, $x^2 = 9$ (equal); at $x = 0$, $x^2 = 0$ (Column B greater) — it varies.

95 Given that $x \geq 7$. Column A: x^2. Column B: 49.

a. The relationship cannot be determined from the given information ✓

b. The two columns are equal

c. Column B is greater

d. Column A is greater

At $x = 7$ they are equal; for $x > 7$, Column A is greater — it varies.

MCQ — More geometry comparisons

96 Column A: the volume of the cube shown. Column B: the volume of a rectangular box $4 \times 5 \times 7$.

a. The relationship cannot be determined from the given information

b. Column A is greater

c. **Column B is greater ✓**

d. The two columns are equal

Cube $= 5^3 = 125$; box $= 4 \times 5 \times 7 = 140$. Column B is greater.

97 Column A: the circumference of a circle with radius 7 cm (use $\pi = \frac{22}{7}$). Column B: the area of a square with side 11 cm.

a. The two columns are equal

b. Column A is greater

c. **Column B is greater ✓**

d. The relationship cannot be determined from the given information

Circumference $= 2 \times \frac{22}{7} \times 7 = 44$; square area $= 121$. Column B is greater.

98 Column A: the sum of the interior angles of the octagon shown. Column B: four times the sum of the interior angles of a triangle.

a. Column B is greater

b. The two columns are equal

c. **Column A is greater ✓**

d. The relationship cannot be determined from the given information

Octagon $= (8 - 2) \times 180 = 1{,}080^\circ$; four triangles $= 4 \times 180 = 720^\circ$. Column A is greater.